

Geneformer

BERT-style encoder

rank-ordered gene tokens

no expression-value embedding

20,271 genes



scGPT

generative transformer

gene tokens + expression

value embeddings

60,694 genes



scFoundation

asymmetric encoder-decoder

continuous expression via

learned auto-discretisation

19,264 genes



identical four-metric geometry screen on the gene-embedding matrix

norm · centroid distance · cosine to centroid · isolation ($k = 10$); outlier if $|z| > 3$ on any metric

410 outliers

188 outliers

164 outliers

all three models

stable under alternative
outlier definitions?

recur in ESM-2
sequence space?

associated with
ClinVar?

Geneformer only

costly to delete vs
matched controls?